

Evolving TTA Challenges Across Scenarios

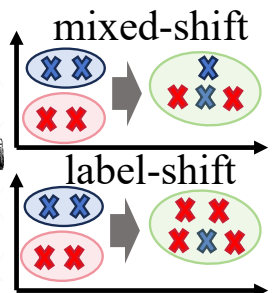
Unlabeled Test-time inputs



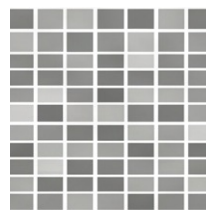
Corrupted
image



Natural
shift



Wild non-i.i.d.
stream



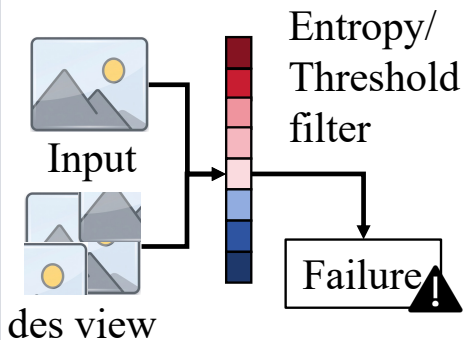
Dense
pixel



Prompt
tokens

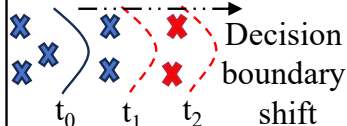
Deployed
online model

(1) Unreliable Adaptation Evidence



Prior methods relying on
scalar probes yield fragile,
inconsistent evidence.

(2) Self-supervised Update Drift



Why prior methods fail:



Static Fisher
anchors cause drift.



Coarse resets drop
vital updates.



EMA/memory
absorbs bias.



Lagging subspaces
delay sudden fixes.

(3) Adaptation Entity Mismatch



Pixel Entity
 $U = (i, s)$



Image Entity
 $U = (i)$



Prompt Entity
 $U = (i, v)$

Prior methods lack
consistent multi-
level loss scaling.